Iván García Mestiza, MSc

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Research Interests: Probability Theory, Stochastic Processes, Applied Probability, Mathematical Finance, Risk Theory.

EDUCATION

- **MSc in Probability and Statistics** 2024 – 2026
  Research Center for Mathematics (CIMAT), Mexico
  Thesis: *Portfolio Optimization under Dynamic Preferences.*
  Thesis Advisor: Dr. Daniel Hernández Hernández
- **BSc in Mathematics** 2019 – 2023
  University of Veracruz (UV), Mexico
  Degree awarded based on academic performance (GPA: 9.97/10).

RESEARCH EXPERIENCE

- **Research Assistant**

  September 2026 – August 2027. Project: *Evolution of Dynamic Preferences and the Heat Equation.* Supervisor: Dr. Daniel Hernández Hernández. Research Center for Mathematics (CIMAT), Mexico.

  - Preparing a research manuscript for publication based on the results of my master's thesis, which investigates the modeling of investors' dynamic preferences through nonnegative solutions of the heat equation and their relationships with conditioned Brownian motions.

  January – December 2022. Project: *Mapping Computational Thinking Research and Practice in Mexico.* Supervisor: Dr. Marcela Quiroz Castellanos. Institute for Artificial Intelligence Research (IIIA), University of Veracruz (UV), Mexico.

  - Research on computational thinking in Mexican education and edition of quarterly newsletters for the Mexican Academy of Computing (AMexComp).

RESEARCH IN PROGRESS

- **Evolution of Dynamic Preferences and the Heat Equation.** Joint work with Dr. Daniel Hernández Hernández, based on my master's thesis. *Manuscript in preparation.*
- **Scale Functions and Ruin Probabilities for Cramér-Lundberg Processes with Matrix-Exponential Claims.** Joint research with Dr. Osvaldo Angtuncio Hernández (UDLAP, Mexico) and Dr. Oscar Peralta (University of Copenhagen, Denmark). We obtained explicit expressions for scale functions and ruin probabilities in the Matrix-Exponential claim size framework. *Manuscript in preparation.*

RESEARCH PRESENTATIONS

1. *Portfolio optimization under dynamic preferences.* At the 59th Congress of the Mexican Mathematical Society (SMM). Mexico, October 2026.
2. *Dynamic preferences through conditioned stochastic processes.* Universidad de las Américas, Puebla (UDLAP). Mexico, October 2026.

3. *A probabilistic perspective on modeling dynamic preferences.* Inter-institutional Seminar on Stochastic Finance. CIMAT-UNAM, Mexico, September 2026.

AWARDS, GRANTS AND RECOGNITIONS

- **Research Assistantship Grant, 2026 – 2027.** Secretariat of Science, Humanities, Technology and Innovation (Secihti), Mexico.
- **Alberto Ruiz Moncayo Award, 2025.** Research Center for Mathematics (CIMAT), Mexico.
  - Awarded for achieving the highest performance in the first-year comprehensive examinations of the M.Sc. in Probability and Statistics program, recognizing the **top-performing student in Probability** of the cohort.
- **Mexican Mathematical Olympiad, 2016 – 2018.**
  - **First Place, National Level**, 2018.
  - **Second Place, National Level**, 2017.
  - First Place, State Level, Veracruz, 2016, 2017, and 2018.
- **National Graduate Studies Scholarship, 2024 – 2026.** Secretariat of Science, Humanities, Technology and Innovation (Secihti), Mexico.
- **Bécalos English Challenge, 2023.**
  - Selected as one of the top-performing English students in Mexico and awarded a scholarship for an advanced English-language study program at ILSC Education Group in Montreal, Canada.
- **Excellence Scholarship, 2021.** Secretariat of Public Education (SEP), Mexico.
- **First Place in University Admission, 2019.** Universidad Veracruzana (UV), Mexico.
  - Highest admission score among applicants to STEM programs at the university.

TEACHING AND OUTREACH

- **Teaching Assistant**

  Semesters 2025-II and 2026-I. Teaching assistant for graduate and undergraduate courses: Statistical Inference I and Statistical Methods. *Course instructor:* Dr. Eloísa Díaz-Francés Murguía. CIMAT-UG, Mexico.

  Semester 2022-II. Teaching assistant for the undergraduate course Real Analysis. *Course instructor:* Dr. Porfirio Toledo Hernández. University of Veracruz (UV), Mexico.

- **Mathematics Olympiad Coach and Judge** 2019 – 2023

  Coach and state-level judge for the Mathematics Olympiad in Veracruz, Mexico.

TECHNICAL SKILLS AND LANGUAGES

- Programming: Python, R, C, C++, JavaScript.
- Spanish (native), English (fluent).